Versatile Water-Floated Nanostructure for Three-Dimensional Nanotransfer Printing

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Abstract

Nanotransfer printing (nTP) provides a cost-effective route to fabricate nanostructures with high spatial fidelity. However, many established processes rely on chemical solvents, adhesive layers, or heat assisted steps. Here, we introduce a water floating based nanotransfer printing approach (WF-nTP), conceptually motivated by hydrographic printing, that enables nanopattern transfer without adhesives, toxic solvents, or high thermal budgets. By tuning the interfacial adhesion between the mold and the patterned layer via plasma assisted surface modification and subsequent etching, the released nanopattern is supported at the air water interface and then deposited onto the target surface. This strategy allows conformal transfer to a broad range of substrates, including curved and low surface energy materials such as optical lenses and plant leaves. WF nTP further supports multilayer assembly of nanostructures for surface enhanced Raman spectroscopy, enabling pesticide detection as a representative demonstration. As an additional application, we fabricated a Pd based hydrogen sensor that shows strong hydrogen sensitivity. These results establish WF nTP as a broadly applicable platform for nanostructure integration in sensing and detection technologies.

Keywords: ***nanotransfer printing, water-floating, plasma etching, SERS***

1. Introduction

Nanotransfer printing (nTP) has become a core strategy for producing nanoscale patterns because it combines low processing cost with high pattern fidelity and broad material compatibility [1]. However, many established nTP schemes still depend on solvent based release steps, adhesive interlayers, or heat assisted processing, which can be incompatible with biological targets and nonplanar geometries [2]. These practical constraints motivate the development of transfer routes that are both environmentally benign and robust across diverse substrates. In this work, we develop water floating nanotransfer printing (WF-nTP), conceptually inspired by hydrographic printing, where the patterned layer is temporarily supported at the air water interface and then delivered to the receiving surface by a simple scooping motion. The key enabler is interfacial adhesion control between the mold and the patterned film via plasma assisted surface modification, which facilitates clean release and substrate independent transfer. As a result, WF-nTP offers a streamlined process window and is particularly effective for three dimensional or otherwise challenging surfaces, expanding the accessible design space for nano enabled devices.

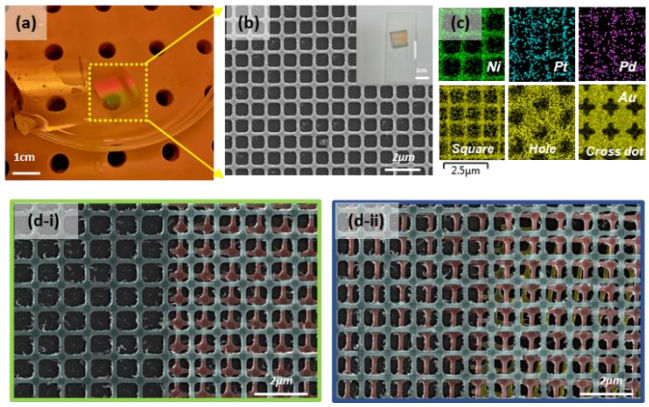

Fig. 1. Various Morphological Analyses of the Floated Nanostructure. (a) Real image of the water-floated nanomesh. (b) SEM image of the Au square mesh transferred onto slide glass. (c) EDS images of various metal meshes and different mesh types. (d) SEM images of the multi-layered mesh on slide glass. The interface image between the 1st and 2nd layers (d-i) and the 2nd and 3rd layers (d-ii), respectively

2. Results and discussion

2.1 Basic Mechanism and Versatility of Floating Pattern

The WF-nTP workflow starts by fabricating a nanopatterned mold on a polyurethane acrylate (PUA) base. A target layer, such as Au, is then deposited onto the mold by electron beam evaporation. To promote clean release, the mold surface is subjected to carefully controlled plasma etching, which lowers the interfacial adhesion between the nanopatterned mold and the deposited film. When the mold is immersed in deionized water, the reduced adhesion allows the patterned nanomaterial to detach and remain supported at the water surface, preparing it for transfer to a wide range of receiving substrates (Fig. 1a). Transfer is completed by a simple scooping step that brings the floating pattern into contact with the target surface. As shown in Fig. 1b, the mesh pattern was delivered onto a glass slide without noticeable structural damage. To demonstrate design and material flexibility, we also transferred alternative mesh geometries, including hole and cross dot patterns, and extended the process beyond Au to Pd and Pt (Fig. 1c). Furthermore, WF-nTP enables layer by layer stacking, allowing the construction of multilayer architectures with added functionality (Fig. 1d).

2.2 Transfer to Three-Dimensional Surfaces

Fig. 2a demonstrates conformal transfer of Au nanomeshes onto optical lenses spanning multiple radii of curvature. To achieve reliable contact on curved geometries, electrical resistance of the transferred meshes was measured while varying the Au thickness (Fig. 2b), which provides an electrical proxy for continuity and effective adhesion. The method also worked on highly textured biological surfaces: Au nanomeshes were deposited onto plant leaves that exhibit micro and nanoscale roughness, yielding

uniform, defect free coverage. Representative photographs of the leaf and tilted SEM views of the mesh on the leaf are shown in Fig. 2c and d. In addition, focused ion beam cross sectioning verified intimate interfacial contact between the mesh and the underlying surface and further revealed a thickness dependent improvement in conformality.

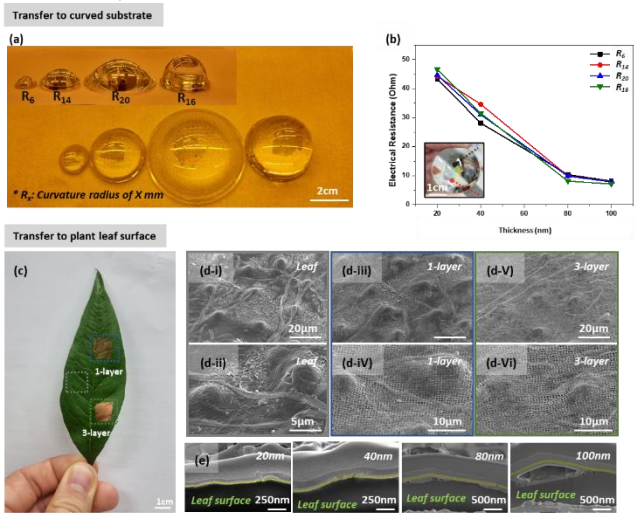

Fig. 2. Transfer Nanomesh to Various 3D Surfaces with morphological analysis. (a) Real image of the nanomesh on the curved optical lens with various curvature radius. (b) Electrical resistance of the nanomesh on the lens according to Au thickness. (c) Real image of the mesh on a leaf with various curved surfaces. (d) SEM images of the raw leaf (d-i and d-ii), 1-layer mesh on the leaf (d-iii and d-iv), and 3-layer mesh on the leaf (d-v and d-vi). (e) Cross-sectional image of the 1-layer mesh on the leaf according to the Au thickness.

2.3 Transfer to Hydrophobic Surfaces

WF-nTP readily operated on wettable targets such as slide glass and the adaxial leaf surface, where the water bath can form stable contact (Fig. 3a). In contrast, transfer to highly hydrophobic substrates, including the abaxial side of a leaf, was hindered by poor wetting at the water interface. To overcome this limitation, we tuned the bath surface tension by introducing ethanol into the water. As shown in Fig. 3b, increasing the ethanol fraction systematically improved wettability on the hydrophobic leaf back surface, enabling reliable transfer. Using this simple adjustment, we

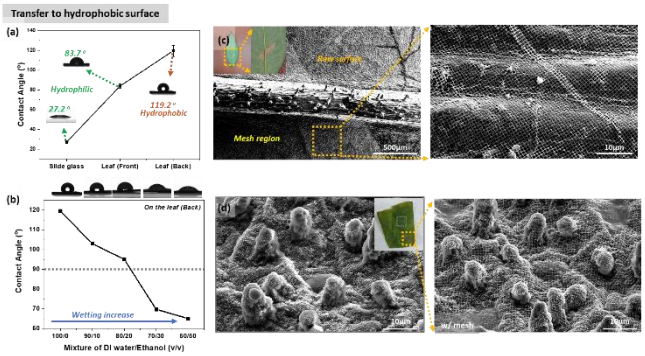

Fig. 3. Transfer to Hydrophobic Surfaces with Morphological Analysis. (a) Different water contact angles on slide glass, the front side, and the back side of the leaf. (b) Contact angle on the back side of the leaf with different volume ratios of DI water and ethanol. (c) SEM images of the mesh on the back side of the leaf. (d) SEM images of the raw lotus surface and transferred mesh on the lotus.

achieved successful deposition on the back side of the leaf (Fig. 3c) and on a lotus leaf, a representative superhydrophobic substrate (Fig. 3d).

2.4 Application to SERS-Based Pesticide Detection and Pd-Based Hydrogen Gas Sensor

The versatility of the WF-nTP process was further validated through two key applications:

Surface-Enhanced Raman Spectroscopy (SERS) Mesh: Using WF nTP, we fabricated multilayer SERS meshes by sequentially stacking the transferred nanomesh layers. FDTD simulations predicted that a seven-layer architecture maximizes local electromagnetic field enhancement. Experimentally, Raman measurements with rhodamine 6G (R6G) verified the expected amplification, enabling detection down to 10^{-6}M. This performance supports non-destructive sensing directly on complex biological surfaces such as leaves and fruits (Fig. 4a). In addition, the platform was applied to pesticide identification, as demonstrated in Fig. 4b.

Pd-Based Hydrogen Gas Sensor: A hydrogen gas sensor was fabricated by integrating a Pd nanomesh with an electrospun TPU fiber substrate. The sensor displayed pronounced responses to sub 1% hydrogen with consistent repeatability (Fig. 4c). It also exhibited high selectivity, showing minimal response to carbon monoxide and nitrogen dioxide (Fig. 4d).

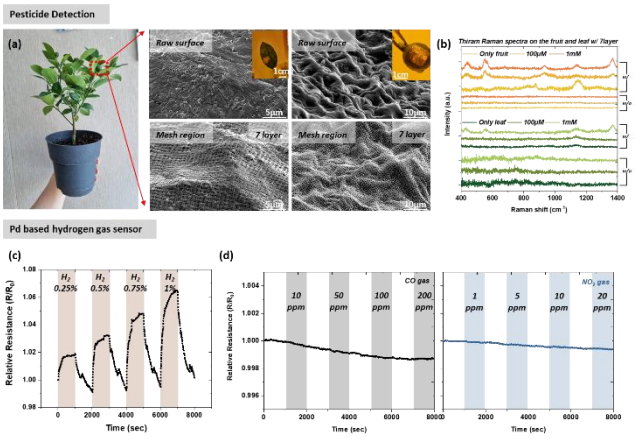

Fig. 4 Demonstration of SERS based pesticide detection and Pd based hydrogen gas senor. (a) Real image of a lemon-orange tree, and SEM images of the leaf and fruit with and without the 7-layer mesh. (b) Thiram Raman spectra on the leaf and fruit at different concentrations of thiram. (c) Response of the hydrogen sensor depending on the gas concentration below 1%. (d) Gas selectivity for different gases, including CO and NO_2, at varying concentrations.

Acknowledgments

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